

- Q1** Consider the poset $(\{3,5,9,15,24,45\})$. Which of the following is correct for the given poset?
- (A) There exists a greatest element and a least element
 (B) There exists a greatest element but not a least element
 (C) There exists a least element but not a greatest element
 (D) There does not exist a greatest element and a least element

- Q2** How many ways are there to place 8 indistinguishable balls into four distinguishable bins?
- (A) 70 (B) 165
 (C) 8C_4 (D) 8P_4

- Q3** How many bit strings of length ten either start with a 1 bit or end with two bits 00?
- (A) 320 (B) 480
 (C) 640 (D) 768

- Q4** Suppose that a connected planar graph has six vertices, each of degree four. Into how many regions is the plane divided by a planar representation of this graph?
- (A) 6 (B) 8
 (C) 12 (D) 20

- Q5** For which values of m and n does the complete bipartite graph $K_{m,n}$ have a Hamilton circuit?
- (A) $m \neq n, m, n \geq 2$
 (B) $m \neq n, m, n \geq 3$
 (C) $m = n, m, n \geq 2$
 (D) $m = n, m, n \geq 3$

- Q6** Which of the following is principal conjunctive normal form for $[(p \vee q) \wedge \neg p \rightarrow \neg q]$?
- (A) $p \vee \neg q$ (B) $p \vee q$
 (C) $\neg p \vee q$ (D) $\neg p \vee \neg q$

- Q7** How many cards must be selected from a standard deck of 52 cards to guarantee that at

least three hearts are present among them?

- (A) 9 (B) 13
 (C) 17 (D) 42

- Q8** Match List-I with List-II :

List-I	List-II
(a) $p \rightarrow q$	(i) $\neg(q \rightarrow \neg p)$
(b) $p \vee q$	(ii) $p \wedge \neg q$
(c) $p \wedge q$	(iii) $\neg p \rightarrow q$
(d) $\neg(p \rightarrow q)$	(iv) $\neg p \vee q$

Choose the correct option from those given below :

- (A) (a)-(ii); (b)-(iii); (c)-(i); (d)-(iv)
 (B) (a)-(ii); (b)-(i); (c)-(iii); (d)-(iv)
 (C) (a)-(iv); (b)-(i); (c)-(iii); (d)-(ii)
 (D) (a)-(iv); (b)-(iii); (c)-(i); (d)-(ii)

- Q9** Find the zero-one matrix of the transitive closure of the relation given by the matrix A :

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

- (A) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$
 (B) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$
 (C) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$
 (D) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

- Q10** Consider an LPP given as

$$\text{Max } Z = 2x_1 - x_2 + 2x_3$$

subject to the constraints

Consider an LPP given as



$$\text{Max } Z = 2x_1 - x_2 + 2x_3$$

subject to the constraints

$$\begin{aligned} 2x_1 + x_2 &\leq 10 \\ x_1 + 2x_2 - 2x_3 &\leq 20 \\ x_1 + 2x_3 &\leq 5 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

What shall be the solution of the LPP after applying first iteration of the Simplex Method?

- (A) $x_1 = \frac{5}{2}, x_2 = 0, x_3 = 0, Z = 5$
 (B) $x_1 = 0, x_2 = 0, x_3 = \frac{5}{2}, Z = 5$
 (C) $x_1 = 0, x_2 = \frac{5}{2}, x_3 = 0, Z = -\frac{5}{2}$
 (D) $x_1 = 0, x_2 = 0, x_3 = 10, Z = 20$

Q11 Which type of addressing mode, less number of memory references are required?

- (A) Immediate (B) Implied
 (C) Register (D) Indexed

Q12 Suppose that the register A and the register K have the bit configuration. Only the three leftmost bits of A are compared with memory words because K has 1's in these positions. Because of its organization, this type of memory is uniquely suited to parallel searches by data association. This type of memory is known as

- (A) RAM
 (B) ROM
 (C) content addressable memory
 (D) secondary memory

Q13 How many different Boolean functions of degree n are there?

- (A) 2^{3^n} (B) 2^{2^n}
 (C) $2^{2^n} - 1$ (D) 2^n

Q14 The fault can be easily diagnosed in the micro-program control unit using diagnostic tools by maintaining the contents of

- (A) flags and counters
 (B) registers and counters
 (C) flags and registers
 (D) flags, registers and counters

Q15 What will be the number of states when a MOD-2 counter is followed by a MOD-5 counter?

- (A) 5 (B) 10
 (C) 15 (D) 20

Q16 For a magnetic disk with concentric circular tracks, the seek latency is not linearly proportional to the seek distance due to
 (A) non-uniform distribution of requests
 (B) arm starting or stopping inertia
 (C) higher capacity of tracks on the periphery of the platter
 (D) More than one of the above

Q17 The parallel bus arbitration technique uses an external priority encoder and a decoder. Suppose, a parallel arbiter has 5 bus arbiters. What will be the size of the priority encoder and decoder respectively?

- (A) $4 \times 2, 2 \times 4$
 (B) $2 \times 4, 4 \times 2$
 (C) $3 \times 8, 8 \times 3$
 (D) $8 \times 3, 3 \times 8$

Q18 Consider the equation $(146)_b + (313)_{b-2} = (246)_8$. Which of the following is the value of b ?

- (A) 8 (B) 7
 (C) 10 (D) 16

Q19 How many address lines and data lines are required to provide a memory capacity of $16 \text{ K} \times 16$?

- (A) 10,4 (B) 16,16
 (C) 14,16 (D) 4,16

Q20 Suppose that a computer program takes 100 seconds of execution time on a computer with multiplication operation responsible for 80 seconds of this time. How much do you have to improve the speed of multiplication operation if you are asked to execute this program four times faster?

(A) 14 times faster
 (B) 15 times faster
 (C) 16 times faster



(D) 17 times faster

- Q21** Consider the following pseudo-code fragment in which an invariant for the loop is " $m * x = p^n$ and $k \geq 0$ " (here, p and n are integer variables that have been initialized):

```
/* Pre-conditions:  $p \geq 1 \wedge n \geq 0$  */
/* Assume that overflow never occurs */
int x = p;
int k = n;
int m = 1;
while (k != 0) {
    if (k is odd) then  $m = m * x$ ;
     $x = x * x$ ;
     $k = \text{floor}(k / 2)$ ; /* equivalent to  $k = k / 2$  */
}
```

Which of the following must be true at the end of the while loop?

- (A) $x = p^n$ (B) $m = p^n$
 (C) $p = x^n$ (D) $p = m^n$

- Q22** Consider the following C-code fragment running on a 32-bit x86 machine:

```
typedef struct {
    union {
        unsigned char a;
        unsigned short b;
    } U;
    unsigned char c;
} S;
S B[10];
S*p=&B[4];
S*q=&B[5];
```

$p \rightarrow U.b = 0x1234$; /* structure S takes 32-bits */

If M is the value of $q - p$ and N is the value of $((\text{int}) \&(p \rightarrow c)) - ((\text{int}) p)$, then (M, N) is

- (A) (1, 1) (B) (3, 2)
 (C) (1, 2) (D) (4, 4)

- Q23** What is the output of the following JAVA program?
 ...

```
public class Good {
    private int m;
    public Good (int m) {this.m = m; }
```

```
public Boolean equals (Good n) { return
n.m == m;
public static void main (String args []) {
    Good m1 = new Good (22);
    Good m2 = new Good (22);
    Object s1 = new Good (22);
    Object s2 = new Good (22);
    System.out.println (m1.equals (m2));
//Line1
    System.out.println (s1.equals (s2));
//Line2
    System.out.println (m1.equals (s2));
//Line3
    System.out.println (s1.equals (m2));
//Line4
}
```

- (A) True, True, False, False
 (B) True, False, True, False
 (C) True, True, False, True
 (D) True, False, False, False

- Q24** Consider the following C++ function $f()$:

```
unsigned int f (unsigned int n) {
    unsigned int b = 0;
    while (n) {
        b += n & 1;
        n >>= 1;
    }
    return b;
}
```

The function $f()$ returns the int that represents the P in the binary representation of positive integer n , where P is

- (A) number of 0's
 (B) number of bits
 (C) number of consecutive 1's
 (D) number of 1's

- Q25** Which of the following statements is/are true?
 P: In a scripting language like JavaScript, types are typically associated with values, not variables.



Q : It is not possible to show images on a web page without the tag of HTML.

Select the correct answer from the options given below :

- (A) P only
- (B) Q only
- (C) Both P and Q
- (D) Neither P nor Q

Q26 Which of the following statements is/are true?

P : An XML document with correct syntax as specified by W3C is called "Well Formed".

Q : An XML document validated against a DTD is both "Well formed" and "Valid".

R : <xml version="1.0" encoding="UTF-8"> is syntactically correct declaration for the version of an XML document.

Select the correct answer from the options given below :

- (A) P and Q only
- (B) P and R only
- (C) Q and R only
- (D) All of P, Q and R

Q27 Consider a raster system with resolution 640 by 480 . What size is frame buffer (in bytes) for this system to store 12 bits per pixel?

- (A) 450 kilobytes
- (B) 500 kilobytes
- (C) 350 kilobytes
- (D) 400 kilobytes

Q28 Consider the following statements regarding 2D transforms in computer graphics:

S1: $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ is a 2×2 matrix that reflects

(mirrors) only 2D point about the X-axis.

S2 : A 2×2 matrix which mirrors any 2D point about the X-axis, is a rotation matrix.

What can you say about the statements S1 and S2?

- (A) Both S1 and S2 are true
- (B) Only S1 is true
- (C) Only S2 is true
- (D) Both S1 and S2 are false

Q29 In the context of 3D computer graphics, which of the following statements is/are true?

P : Orthographic transformations keep parallel lines parallel.

Q : Orthographic transformations are affine transformations.

Select the correct answer from the options given below :

- (A) Both P and Q
- (B) Neither P nor Q
- (C) Only P
- (D) Only Q

Q30 Using the phong reflectance model, the strength of the specular highlight is determined by the angle between

- (A) the view vector and the normal vector
- (B) the light vector and the normal vector
- (C) the light vector and the reflected vector
- (D) the reflected vector and the view vector

Q31 Which of the following has same expressive power with regard to relational query language?

- (a) Relational algebra and domain relational calculus
- (b) Relational algebra and tuples relational calculus
- (c) Relational algebra and domain relational calculus restricted to safe expression
- (d) Relational algebra and tuples relational calculus restricted to safe expression

- (A) (a) and (b) only
- (B) (c) and (d) only
- (C) (a) and (c) only
- (D) (b) and (d) only

Q32 Which of the following features is supported in the relational database model?

- (A) Complex data-types
- (B) Multivalued attributes
- (C) Associations with multiplicities
- (D) Generalization relationships

Q33 With respect to relational algebra, which of the following operations are included for mathematical set theory?

- (a) Join
- (b) Intersection



(c) Cartesian product

(d) Project

(A) (a) and (d)

(B) (b) and (c)

(C) (c) and (d)

(D) (b) and (d)

Q34 In relational databases, if relation R is in BCNF, then which of the following is true about relation R ?

(A) R is in 4NF

(B) R is not in 1NF

(C) R is in 2NF and not in 3NF

(D) R is in 2NF and 3NF

Q35 Which of the following key constraints is required for functioning of foreign key in the context of relational databases?

(A) Unique key

(B) Primary key

(C) Candidate key

(D) Check key

Q36 In relational database management, which of the following is/are property/properties of candidate key?

P : Uniqueness

Q : Irreducibility

(A) P only

(B) Q only

(C) Both P and Q

(D) Neither P nor Q

Q37 Which of the following statements are DML statements?

(a) Update [tablename]

Set [columnname] = VALUE

(b) Delete [tablename]

(c) Select * from [tablename]

(A) (a) and (b)

(B) (a) and (d)

(C) (a), (b) and (c)

(D) (b) and (c)

Q38 Hadoop (a big data tool) works with number of related tools. Choose from the following, the common tools included into Hadoop :

(A) MySQL, Google API and Map reduce

(B) Map reduce, Scala and Hummer

(C) Map reduce, H Base and Hive

(D) Map reduce, Hummer and Heron

Q39 The following table has two attributes Employee_id and Manager_id, where Employee_id is a primary key and manager_id is a foreign key referencing Employee_id with an on-delete cascade :

Employee_id	Manager_id
20	40
25	40
30	35
35	20
40	45
45	25

On deleting the tuple (20, 40), the set of other tuples that must be deleted to maintain the referential integrity of the table is:

(A) (30, 35) only

(B) (30, 35) and (35, 20) only

(C) (35, 20) only

(D) (40, 45) and (25, 40) only

Q40 K-mean clustering algorithm has clustered the given 8 observations into 3 clusters after 1st iteration as follows:

C1 : {(3, 3), (5, 5), (7, 7)}

C2 : {(0, 6), (6, 0), (3, 0)}

C3 : {(8, 8), (4, 4)}

What will be the Manhattan distance for observation (4, 4) from cluster centroid C1 in second iteration?

(A) 2

(B) $\sqrt{2}$

(C) 0

(D) 18

Q41 Consider a disk system with 100 cylinders. The requests to access the cylinders occur in the following sequence :

4, 34, 10, 7, 19, 73, 2, 15, 6, 20

Assuming that the head is currently at cylinder 50, what is the time taken to satisfy all requests if it takes 1 ms to move from the cylinder to



adjacent one and the shortest seek time first policy is used?

- (A) 357 ms (B) 238 ms
(C) 276 ms (D) 119 ms

Q42 Match List-I with List-II :

List-I	List-II
(a) Disk	(i) Thread
(b) CPU	(ii) Signal
(c) Memory	(iii) File system
(d) Interrupt	(iv) Virtual address space

Choose the correct option from those given below :

- (A) (a)-(i); (b)-(ii); (c)-(iii); (d)-(iv)
(B) (a)-(iii); (b)-(i); (c)-(iv); (d)-(ii)
(C) (a)-(ii); (b)-(i); (c)-(iv); (d)-(iii)
(D) (a)-(ii); (b)-(iv); (c)-(iii); (d)-(i)

Q43 Which of the following are NOT shared by the threads of the same process?

- (a) Stack
(b) Registers
(c) Address space
(d) Message queue
(A) (a) and (d)
(B) (b) and (c)
(C) (a) and (b)
(D) (a), (b) and (c)

Q44 Consider that a process has been allocated 3 frames and has a sequence of page referencing as 1, 2, 1, 3, 7, 4, 5, 6, 3, 1.

What shall be the difference in page faults for the above string using the algorithms of LRU and optimal page replacement for referencing the string?

- (A) 2 (B) 0
(C) 1 (D) 3

Q45 Consider three CPU intensive processes, which require 10, 20 and 30 units of time and arrive at times 0, 2 and 6 respectively. How many context switches are needed if the operating system implements a shortest remaining time first

scheduling algorithm? Do not count the context switches at time zero and at the end.

- (A) 4 (B) 2
(C) 3 (D) 1

Q46 At a particular time of computation, the value of a counting semaphore is 7. Then 20 P (wait) operations and 15 V (signal) operations are completed on this semaphore. What is the resulting value of the semaphore?

- (A) 28 (B) 12
(C) 2 (D) 42

Q47 The minimum number of page frames that must be allocated to a running process in a virtual memory environment is determined by

- (A) page size
(B) physical size of memory
(C) the instruction set architecture
(D) number of processes in memory

Q48 A computer has six tape drives with (n) processes competing for them. Each process may need two drives. What is the maximum value of (n) for the system to be deadlock free?

- (A) 5 (B) 4
(C) 3 (D) 6

Q49 A processor can support a maximum memory of (4 GB) where memory is word addressable and a word is 2 bytes. What will be the size of the address bus of the processor?

- (A) At least 28 bits
(B) At least 2 bytes
(C) At least 31 bits
(D) Minimum 4 bytes

Q50 Which of the following UNIX/Linux pipes will count the number of lines in all the files having .c and h as their extension in the current working directory?

- (A) $\text{cat } * \cdot \text{ch} \mid \text{wc} - 1$
(B) $\text{cat } * \cdot [\text{c} - \text{h}] \mid \text{wc} - 1$
(C) $\text{cat } * \cdot [\text{ch}], 1 \text{ s} - 1$
(D) $\text{cat } *.[\text{ch}], \text{wc} - 1$

Q51



Which of the following statements is/are true?

P : In software engineering, defects that are discovered earlier are more expensive to fix.

Q : A software design is said to be a good design, if the components are strongly cohesive and weakly coupled.

Select the correct answer from the options given below :

- (A) P only
- (B) Q only
- (C) P and Q
- (D) Neither P nor Q

- Q52** The M components in MVC are responsible for
- (A) user interface
 - (B) security of the system
 - (C) business logic and domain objects
 - (D) translating between user interface actions/events and operations on the domain objects

- Q53** In the context of software testing, which of the following statements is/are NOT correct?

P : A minimal test set that achieves 100 % path coverage will also achieve 100 % statement coverage.

Q : A minimal test set that achieves 100 % path coverage will generally detect more faults than one that achieves 100 % statement coverage.

R : A minimal test set that achieves 100 % statement coverage will generally detect more faults than one that achieves 100 % branch coverage.

- (A) R only
- (B) Q only
- (C) P and Q only
- (D) Q and R only

- Q54** Software reuse is
- (A) the process of analysing software with the objective of recovering its design and specification
 - (B) the process of using existing software artifacts and knowledge to build new software
 - (C) concerned with reimplementing legacy system to make them more maintainable

(D) the process of analysing software to create a representation of a higher level of abstraction and breaking software down into its parts to see how it works

- Q55** Which of the following terms best describes Git?
- (A) Issue Tracking System
 - (B) Integrated Development Environment
 - (C) Distributed Version Control System
 - (D) Web-based Repository Hosting Service

- Q56** A Web application and its support environment has not been fully fortified against attack. Web engineers estimate that the likelihood of repelling an attack is only 30 percent. The application does not contain sensitive or controversial information, so the threat probability is 25 percent. What is the integrity of the web application?
- (A) 0.625
 - (B) 0.725
 - (C) 0.775
 - (D) 0.825

- Q57** Match List-I with List-II :

List-I (Software Process Models)	List-II (Software Systems)
(a) Waterfall model	(i) e-business software that starts with only the basic functionalities and then moves on to advanced features
(b) Incremental development	(ii) An inventory control system for a supermarket to be developed within three months
(c) Prototyping	(iii) A virtual reality system for simulating vehicle navigation in a highway
(d) RAD	(iv) Automate the manual system for student record maintenance in a school

Choose the correct option from those given below :



- (A) (a)-(ii); (b)-(iv); (c)-(i); (d)-(iii)
 (B) (a)-(i); (b)-(iii); (c)-(iv); (d)-(ii)
 (C) (a)-(iii); (b)-(ii); (c)-(iv); (d)-(i)
 (D) (a)-(iv); (b)-(i); (c)-(iii); (d)-(ii)

Q58 Software products need adaptive maintenance for which of the following reasons?

- (A) To rectify bugs observed while the system is in use
 (B) When the customers need the product to run on new platforms
 (C) To support the new features that users want it to support
 (D) To overcome wear and tear caused by the repeated use of the software

Q59 Which of the following are the primary objectives of risk monitoring in software project tracking?

P : To assess whether predicted risks do, in fact, occur

Q : To ensure that risk aversion steps defined for the risk are being properly applied

R : To collect information that can be used for future risk analysis

- (A) Only P and Q (B) Only P and R
 (C) Only Q and R (D) All of P,Q,R

Q60 Software validation mainly checks for inconsistencies between

- (A) use cases and user requirements
 (B) implementation and system design blueprints
 (C) detailed specifications and user requirements
 (D) functional specifications and use cases

Q61 Match List-I with List-II :

List-I	List-II
(a) Prim's algorithm	(i) $O(V^3 \log V)$
(b) Dijkstra's algorithm	(ii) $O(VE^2)$
(c) Faster all-pairs shortest path	(iii) $O(E \lg V)$
(d) Edmonds-Karp algorithm	(iv) $O(V^2)$

Choose the correct option from those given below :

- (A) (a)-(ii); (b)-(iv); (c)-(i); (d)-(iii)

- (B) (a)-(iii); (b)-(iv); (c)-(i); (d)-(ii)
 (C) (a)-(ii); (b)-(i); (c)-(iv); (d)-(iii)
 (D) (a)-(iii); (b)-(i); (c)-(iv); (d)-(ii)

Q62 There are many sorting algorithms based on comparison. The running time of heapsort algorithm is $O(n \lg n)$. Like P, but unlike Q, heapsort sorts in place where (P, Q) is equal to
 (A) Merge sort, Quick sort
 (B) Quick sort, insertion sort
 (C) Insertion sort, Quick sort
 (D) Insertion sort, Merge sort

Q63 Consider the Euler's phi function given by $\phi(n) = n \prod \left(1 - \frac{1}{p}\right)$ where p runs over all the primes dividing n . What is the value of $\phi(45)$?

- (A) 3 (B) 12
 (C) 6 (D) 24

Q64 Which of the following is best running time to sort n integers in the range 0 to $n^2 - 1$?

- (A) $O(\lg n)$ (B) $O(n)$
 (C) $O(n \lg n)$ (D) $O(n^2)$

Q65 Which of the following is application of depth-first search?

- (A) Only topological sort
 (B) Only strongly connected components
 (C) Both topological sort and strongly connected components
 (D) Neither topological sort nor strongly connected components

Q66 Consider double hashing of the form

$$h(k, i) = (h_1(k) + i h_2(k)) \bmod m$$

where $h_1(k) = k \bmod m$

$$h_2(k) = 1 + (k \bmod n)$$

where $n = m - 1$ and $m = 701$

For $k = 123456$, what is the difference between first and second probes in terms of slots?



- (A) 255 (B) 256
(C) 257 (D) 258

Q67 Consider the complexity class $CO - NP$ as the set of languages L such that $\bar{L} \in NP$, and the following two statements :

$S_1 : P \subseteq CO - NP$

$S_2 : \text{If } NP \neq CO - NP, \text{ then } P \neq NP$

Which of the following is/are correct?

- (A) Only S_1
(B) Only S_2
(C) Both S_1 and S_2
(D) Neither S_1 nor S_2

Q68 Consider the following steps :

S_1 : Characterize the structure of an optimal solution

S_2 : Compute the value of an optimal solution in bottom-up fashion

Which of the step(s) is/are common to both dynamic programming and greedy algorithms?

- (A) Only S_1
(B) Only S_2
(C) Both S_1 and S_2
(D) Neither S_1 nor S_2

Q69 Consider the following properties with respect to a flow network $G = (V, E)$ in which a flow is a real-valued function $f : V \times V \rightarrow R$:

P_1 : For all $u, v \in V, f(u, v) = -f(v, u)$

$P_2 : \sum_{v \in V} f(u, v) = 0$ for all $u \in V$

Which one of the following is/are correct?

- (A) Only P_1
(B) Only P_2
(C) Both P_1 and P_2
(D) Neither P_1 nor P_2

Q70 Consider the following statements :

S_1 : For any integer $n > 1, a^{\phi(n)} \equiv 1 \pmod{n}$ for all $a \in Z_n^*$, where $\phi(n)$ is Euler's phi function.

S_2 : If p is prime, then $a^p \equiv 1 \pmod{p}$ for all $a \in Z_p^*$.

Which one of the following is/are correct?

- (A) Only S_1
(B) Only S_2
(C) Both S_1 and S_2
(D) Neither S_1 nor S_2

Q71 Which data structure is used by the compiler for managing variables and their attributes?

- (A) Binary tree (B) Link list
(C) Symbol table (D) Parse table

Q72 On translating the expression given below into quadruple representation, how many operations are required?

$(i*j)+(e+f)*(a*b+c)$

- (A) 5 (B) 6
(C) 3 (D) 7

Q73 Replacing the expression $4 * 2.14$ by 8.56 is known as

- (A) constant folding
(B) induction variable
(C) strength reduction
(D) code reduction

Q74 Shift-reduce parser consists of

- (a) input buffer
(b) stack
(c) parse table

Choose the correct option from those give below :

- (A) (a) and (b) only
(B) (a) and (c) only
(C) (c) only
(D) (a), (b) and (c)

Q75 How many states are there in a minimum state automata equivalent to regular expression given below?

Regular expression is $a * b(a + b)$

- (A) 1 (B) 2
(C) 3 (D) 4

Q76 Match List-I with List-II :

where L_1 : Regular language

L_2 : Context-free language

L_3 : Recursive language

L_4 : Recursively enumerable language



List-I	List-II
(a) $\bar{L}_3 \cup L_4$	(i) Context-free language
(b) $\bar{L}_2 \cup L_3$	(ii) Recursively enumerable language
(c) $L_1^* \cap L_2$	(iii) Recursive language

Choose the correct option from those given below :

- (A) (a)-(ii); (b)-(i); (c)-(iii)
 (B) (a)-(ii); (b)-(iii); (c)-(i)
 (C) (a)-(iii); (b)-(i); (c)-(ii)
 (D) (a)-(i); (b)-(ii); (c)-(iii)

Q77 How can the decision algorithm be constructed for deciding whether context-free language L is finite?

- (a) By constructing redundant CFG G in CNF generating language L
 (b) By constructing non-redundant CFG G in CNF generating language L
 (c) By constructing non-redundant CFG G in CNF generating language $L - \{\wedge\}$ ($\wedge$ stands for null)

Which of the following is correct?

- (A) (a) only
 (B) (b) only
 (C) (c) only
 (D) None of (a), (b) and (c)

Q78 Consider the following grammar :

$$S \rightarrow XY$$

$$X \rightarrow YaY \mid a \text{ and } Y \rightarrow bbX$$

Which of the following statements is/are true about the above grammar?

- (a) Strings produced by the grammar can have consecutive three a 's.
 (b) Every string produced by the grammar have alternate a and b .
 (c) Every string produced by the grammar have at least two a 's.
 (d) Every string produced by the grammar have b 's in multiple of 2.

- (A) (a) only
 (B) (b) and (c) only
 (C) (d) only
 (D) (c) and (d) only

Q79 Which of the following problems is/are decidable problem(s) (recursively enumerable) on turing machine M ?

- (a) G is a CFG with $L(G) = \phi$
 (b) There exist two TMs M_1 and M_2 such that $L(M) \subseteq \{L(M_1) \cup L(M_2)\}$ = language of all TMs
 (c) M is a TM that accepts w using at most $2^{|w|}$ cells of tape
 (A) (a) and (b) only
 (B) (a) only
 (C) (a), (b) and (c)
 (D) (c) only

Q80 For a statement

A language $L \subseteq \Sigma^*$ is recursive if there exists some turing machine M .

Which of the following conditions is satisfied for any string w ?

- (A) If $w \in L$, then M accepts w and M will not halt
 (B) If $w \notin L$, then M accepts w and M will halt by reaching at final state
 (C) If $w \notin L$, then M halts without reaching to acceptable state
 (D) If $w \in L$, then M halts without reaching to an acceptable state

Q81 A fully connected network topology is a topology in which there is a direct link between all pairs of nodes. Given a fully connected network with n nodes, the number of direct links as a function of n can be expressed as

- (A) $\frac{n(n+1)}{2}$ (B) $\frac{(n+1)}{2}$
 (C) $\frac{n}{2}$ (D) $\frac{n(n-1)}{2}$

Q82 What is the name of the protocol that allows a client to send a broadcast message with its MAC address and receive an IP address in reply?

- (A) ARP (B) DNS



(C) RARP

(D) ICMP

Q83 You are designing a link layer protocol for a link with a bandwidth of 1 Gbps (10^9 bits/second) over a fiber link with a length of 800 km.

Assume the speed of light in this medium is 200000 km/second. What is the propagation delay in this link?

- (A) 1 millisecond (B) 2 milliseconds
(C) 3 milliseconds (D) 4 milliseconds

Q84 You need 500 subnets, each with about 100 usable host addresses per subnet. What network mask will you assign using a class B network address?

- (A) 255.255.255.252
(B) 255.255.255.128
(C) 255.255.255.0
(D) 255.255.254.0

Q85 Consider the following two statements with respect to IPv4 in computer networking :

P: The loopback (IP) address is a member of the class B network.

Q: The loopback (IP) address is used to send a packet from the host to itself.

What can you say about the statements P and Q?

- (A) P-True; Q-False
(B) P-False; Q-True
(C) P-True; Q-True
(D) P-False; Q-False

Q86 In the TCP/IP model, encryption and decryption are functions of the ____ layer.

- (A) Data link (B) Network
(C) Transport (D) Application

Q87 Which of the following statements is/are true with regard to various layers in the Internet stack?

P: At the link layer, a packet of transmitted information is called a frame.

Q: At the network layer, a packet of transmitted information is called a segment.

- (A) P only
(B) Q only

(C) P and Q

(D) Neither P nor Q

Q88 What percentage % of the IPv4, IP address space do all class C addresses consume?

- (A) 12.5% (B) 25%
(C) 37.5% (D) 50%

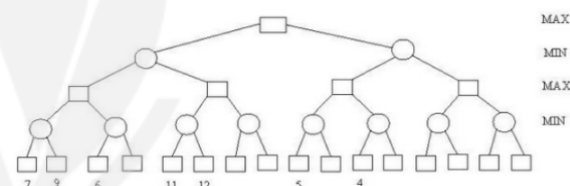
Q89 The ability to inject packets into the Internet with a false source address is known as

- (A) Man-in-the-middle attack
(B) IP phishing
(C) IP sniffing
(D) IP spoofing

Q90 The RSA encryption algorithm also works in reverse, that is, you can encrypt a message with the private key and decrypt it using the public key. This property is used in

- (A) intrusion detection systems
(B) digital signatures
(C) data compression
(D) certification

Q91 Consider the game tree given below :



Here  and  represents MIN and MAX nodes respectively. The value of the root node of the game tree is

- (A) 4 (B) 7
(C) 11 (D) 12

Q92 Match List-I with List-II :

List I	List II
(a) Greedy best-first	(i) Minimal cost (p) + h(p)
(b) Lowest cost-first	(ii) Minimal h(p)
(c) A* Algorithm	(iii) Minimal cost (p)

Choose the correct option from those given below :

- (A) (a)-(i); (b)-(ii); (c)-(iii)
(B) (a)-(iii); (b)-(ii); (c)-(i)



(C) (a)-(i); (b)-(iii); (c)-(ii)

(D) (a)-(ii); (b)-(iii); (c)-(i)

Q93 The STRIPS representation is

(A) a feature-centric representation

(B) an action-centric representation

(C) a combination of feature-centric and action-centric representations

(D) a hierarchical feature-centric representation

Q94 A fuzzy conjunction operator denoted as $t(x,y)$ and a fuzzy disjunction operator denoted as $s(x,y)$ form a dual pair if they satisfy the condition :

(A) $t(x,y) = 1 - s(x,y)$

(B) $t(x,y) = s(1-x, 1-y)$

(C) $t(x,y) = 1 - s(1-x, 1-y)$

(D) $t(x,y) = s(1+x, 1+y)$

Q95 Let A_{α_0} denotes the α -cut of a fuzzy set A at α_0 . If $\alpha_1 < \alpha_2$, then

(A) $A_{\alpha_1} \supseteq A_{\alpha_2}$

(B) $A_{\alpha_1} \supset A_{\alpha_2}$

(C) $A_{\alpha_1} \subseteq A_{\alpha_2}$

(D) $A_{\alpha_1} \subset A_{\alpha_2}$

Q96 Consider the following methods :

M1: Mean of maximum

M2: Centre of area

M3: Height method

Which of the following is/are defuzzification method(s)?

(A) Only M2 (B) Only M1 and M2

(C) Only M2 and M3 (D) M1, M2, and M3

Q97 Consider the following :

(a) Evolution

(b) Selection

(c) Reproduction

(d) Mutation

Which of the following are found in genetic algorithms?

(A) (b), (c) and (d) only

(B) (b) and (d) only

(C) (a), (b), (c) and (d)

(D) (a), (b) and (d) only

Q98 Which of the following is an example of unsupervised neural network?

(A) Back-propagation network

(B) Hebb network

(C) Associative memory network

(D) Self-organizing feature map

Q99 The value of the derivative of the Sigmoid function given by

$$f(x) = \frac{1}{1+e^{-2x}}$$

at $x=0$ is

(A) 0

(B) $\frac{1}{2}$

(C) $\frac{1}{4}$

(D) ∞

Q100 Reinforcement learning can be formalized in terms of _____ in which the agent initially only knows the set of possible _____ and the set of possible actions.

(A) Markov decision processes, objects

(B) Hidden states, objects

(C) Markov's decision process, states

(D) objects, states



Answer Key

Q1 (D)
Q2 (B)
Q3 (C)
Q4 (B)
Q5 (C)
Q6 (A)
Q7 (D)
Q8 (D)
Q9 (A)
Q10 (B)
Q11 (C)
Q12 (C)
Q13 (B)
Q14 (D)
Q15 (B)
Q16 (B)
Q17 (D)
Q18 (B)
Q19 (C)
Q20 (C)
Q21 (B)
Q22 (C)
Q23 (D)
Q24 (D)
Q25 (A)
Q26 (A)
Q27 (A)
Q28 (B)
Q29 (A)
Q30 (D)
Q31 (B)

Q32 (C)
Q33 (B)
Q34 (D)
Q35 (B)
Q36 (C)
Q37 (C)
Q38 (C)
Q39 (B)
Q40 (A)
Q41 (D)
Q42 (B)
Q43 (C)
Q44 (A)
Q45 (B)
Q46 (C)
Q47 (C)
Q48 (A)
Q49 (C)
Q50 (D)
Q51 (B)
Q52 (C)
Q53 (A)
Q54 (B)
Q55 (C)
Q56 (D)
Q57 (D)
Q58 (B)
Q59 (D)
Q60 (C)
Q61 (B)
Q62 (D)



Q63 (D)
Q64 (B)
Q65 (C)
Q66 (C)
Q67 (C)
Q68 (A)
Q69 (C)
Q70 (A)
Q71 (C)
Q72 (B)
Q73 (A)
Q74 (D)
Q75 (C)
Q76 (B)
Q77 (C)
Q78 (D)
Q79 (C)
Q80 (C)
Q81 (D)

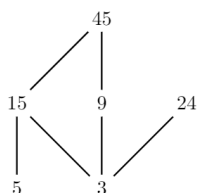
Q82 (C)
Q83 (D)
Q84 (B)
Q85 (B)
Q86 (D)
Q87 (A)
Q88 (A)
Q89 (D)
Q90 (B)
Q91 (B)
Q92 (D)
Q93 (B)
Q94 (C)
Q95 (A)
Q96 (D)
Q97 (C)
Q98 (D)
Q99 (B)
Q100 (C)



Hints & Solutions

Q1 Text Solution:

Correct option is D.



There are two maximal elements 24 and 45.

There are two minimal elements 5 and 3.

So there is no greatest and least element.

Q2 Text Solution:

Correct option is B.

8 indistinguishable balls $\Rightarrow$ only Number of Balls Matter (which balls does not matter)

4 distinguishable bins $\Rightarrow$ Bins Matter

Number of Ways to place 8 indistinguishable balls into 4 distinguishable bins

$$= {}^{8+4-1}C_{4-1}$$

$$= {}^{11}C_3$$

$$= 11!/(8!3!)$$

$$= 11 \cdot 10 \cdot 9 / (3 \cdot 2 \cdot 1)$$

$$= 165$$

Q3 Text Solution:

Correct option is C.

The problem can be divided into two parts,

Part 1 - No. of bit strings of length 10 that start with bit 1.

So, the strings will be of the form

1 _ _ _ _ _

which means 9 empty places to fill with either 0 or 1 hence there can be 2^9 ways.

Part 2 - No. of bit strings of length 10 that end with two bit zeros.

Now, we have already counted the strings that end with two bit zeros in the part 1, but all those strings were starting with a bit 1. So, we still haven't counted the bit strings that start with a zero, and end with two zeroes. So, such strings will be of the form

0 _ _ _ _ _ 0 0

which means 7 empty places to fill with either 0 or 1, hence 2^7 ways.

Hence, total no. of ways = Part 1 + Part 2

$$= 2^9 + 2^7$$

$$= 2^7(2^2 + 1)$$

$$= 2^7(5) = 640$$

Q4 Text Solution:

Correct option is B.

sum of degree of vertices = $2 \times$ number of edges (handshaking theorem)

$$= 4 + 4 + 4 + 4 + 4 + 4 = 2e$$

$$e = 12$$

Also according to Euler's formula,

$$r = e - v + 2$$

$$= r = 12 - 6 + 2$$

$$r = 8$$

Q5 Text Solution:

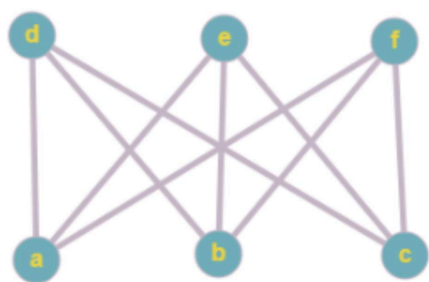
The correct option is C.

Hamiltonian Circuit: In this circuit, every vertex is visited only once. It must start and end with the same vertex.

In a complete bipartite graph $K_{m,n}$, when $m = n$, it has a Hamiltonian circuit.

Consider $m = n = 3$





Hamiltonian circuit possible in this is = {d, a, e, c, f, b, d}

Q6 Text Solution:

The correct option is A.

$$[(p \vee q) \wedge p \rightarrow \neg q]$$

$$= [((p \vee q) \wedge \neg p) \rightarrow \neg q]$$

$$= [((p \wedge \neg p) \vee (q \wedge \neg p)) \rightarrow \neg q] \quad (\text{Using distributive law})$$

$$= [(p \vee (q \wedge \neg p)) \rightarrow \neg q]$$

$$= [\neg(q \wedge \neg p) \vee \neg q]$$

$$= [\neg q \vee p \vee \neg q]$$

$$= [p \vee \neg q] \quad (\text{Using de Morgan's law})$$

Q7 Text Solution:

The correct option is D.

In this, the worst case will happen as follows:-

1st $13 \times 3 = 39$ cards that we pick up turn out to be any one of diamonds or spades or clubs but not hearts.

When we pick the next 3 cards they are guaranteed to be hearts. (.. we have already picked cards from all other suits)

We need to pick $13+13+13+3 = 42$ cards to guarantee that there are at least 3 cards of hearts in the selected cards.

Q8 Text Solution:

The correct option is D.

(a) $p \rightarrow q$, we can write $\neg p \vee q$, so the option is (iv)

(b) $p \vee q = 1(\neg p) \vee q = 1p \rightarrow q$, so the option is (iii)

(c) $p \wedge q = q \wedge p = 1(\neg q \vee \neg p) = 1(q \rightarrow p)$, so the option is (i)

(d) $p \wedge \neg q = 1(\neg p \vee q) = 1(p \rightarrow q)$, so the option is (ii)

Q9 Text Solution:

The correct option is A.

Let the elements be a,b,c on which the relation is defined.

Then from the matrix we can say that the relation given is {(a,a), (b,b), (c,a), (c,b), (a,c)}

Now, what are the minimum number of ordered pairs that we should add to the given relation that it becomes transitive relation?

Using Warshall's Algorithm we can see that (c,c) and (a,b) are missing.

After adding them to the relation and making the matrix again the matrix will become

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Q10 Text Solution:

The correct option is B.

We maximize:

We maximize:

$$Z = 2x_1 - x_2 + 2x_3$$

Adding Slack Variables s_1, s_2, s_3

Subject to:

$$2x_1 + x_2 + s_1 = 10$$

$$x_1 + 2x_2 - 2x_3 + s_2 = 20$$

$$x_1 + 2x_3 + s_3 = 5$$

Tableau Initialization

The initial simplex tableau will be

BV(B)	x_1	x_2	x_3	s_1	s_2	s_3	RHS
s_1	2	1	0	1	0	0	10
s_2	1	2	-2	0	1	0	20
s_3	1	0	2	0	0	1	5

BV(B)	x_1	x_2	x_3	s_1	s_2	s_3	RHS
Z	-2	1	-2	0	0	0	0

Identify pivot column

The most negative entry in the Z-row indicates the entering variable:

x_3 with coefficient -2.

Identify pivot Row

Compute the ratio RHS/pivot Column for non-negative entries in the pivot column:



For s1: $10/0 \rightarrow$ undefined(ignore)

For s2: $20/(-2) \rightarrow$ Negative (ignore)

For s3: $5/2 = 2.5$

The pivot row is s3.

Perform pivoting

The pivot element is 2 (from s3,x3). Normalize the pivot row by dividing it by 2.

Updated table:

BV(B)	x1	x2	x3	s1	s2	s3	RHS
s1	2	1	0	1	0	0	10
s2	1	2	0	0	1	1	10
x3	0.5	0	1	0	0	0.5	2.5
Z	-1	1	0	0	0	1	5

Interpret the Solution:

Basic Variables: $x1=0, x2=0, x3=2.5$

Objective value: $Z = 5$.

Q11 Text Solution:

Explanation:

Correct Answer: C — Register Addressing

Register addressing uses CPU registers, not memory.

Accessing a register takes zero memory references.

Therefore, it is the fastest and requires the least number of memory accesses.

Other Options:

Immediate $\rightarrow$ instruction contains the constant itself, but still needs to fetch the instruction from memory.

Implied $\rightarrow$ depends on instruction; not always minimal memory use.

Indexed $\rightarrow$ needs memory access to compute effective address.

Q12 Text Solution:

The correct option is C.

Associative memory: A type of computer memory from which items may be retrieved by matching some part of their content, rather than

by specifying their address (hence also called associative storage or Content-addressable memory (CAM)).

Q13 Text Solution:

Explanation:

Formula: Number of Boolean functions $= 2^{2^n}$

A Boolean function with n variables has 2^n possible input combinations. For each combination, the output can be 0 or 1 $\Rightarrow$ Thus, total functions $= 2^{2^n}$

Some Examples:

1. If $n = 1$ then $2^{2^1} = 2^2 = 4$ functions

2. If $n = 2$ then $2^{2^2} = 2^4 = 16$ functions

3. If $n = 3$ then $2^{2^3} = 2^8 = 256$ functions

The Final Answer is Option B.

Q14 Text Solution:

The correct option is D.

The fault can be easily diagnosed in the microprogram control unit using diagnostics tools by maintaining the contents of flags, registers and counters.

Q15 Text Solution:

The correct option is B.

When the mod- x counter is followed by the mod- y counter then the resultant counter is always the mod- $(x*y)$ counter.

Simply multiply MOD value $2*5 = 10$

Q16 Text Solution:

The correct option is B.

Whenever head moves from one track to other then its speed and direction changes, which is noting but change in motion or the case of inertia.

Q17 Text Solution:

It is given that the parallel arbiter has 5 bus arbiters. Hence, the number of bits required to provide 5 bus arbiters is 3 bits (Since, $2^3 = 8$). Now encoder is a type of combinational circuit that consists of 2^n and n input lines and output lines respectively. i.e., $2^n \times n$ encoder. Now, $n=3$



bits. Therefore, the size of the encoder is 8×3 .
Similarly, the decoder is a combinational circuit that consists of n -input lines and 2^n output lines, i.e., $n \times 2^n$ decoder. So the size of the decoder is 3×8 .

Q18 Text Solution:

The correct option is B.

First, we substitute and check the result $\Rightarrow b = 7$

$$\Rightarrow (146)_7 + (313)_{7-2} = (246)_8$$

$$\Rightarrow \text{LHS} = (146)_7 + (317)_{7-2}$$

$$\Rightarrow (146)_7 = 1 \times 7^2 + 4 \times 7^1 + 6 \times 7^0 = 49 + 28 + 6 = (83)_{10}$$

$$\Rightarrow (313)_5 = 3 \times 5^2 + 1 \times 5 + 3 \times 5^0 = 75 + 5 + 3 = (83)_{10}$$

$$\Rightarrow \text{LHS} = (83)_{10} + (83)_{10} = (166)_{10}$$

$$\Rightarrow \text{RHS} = (246)_8$$

$$\Rightarrow 2 \times 8^2 + 4 \times 8^1 + 6 \times 8^0$$

$$\Rightarrow 128 + 32 + 6 = (166)_{10}$$

LHS = RHS equals only if b is 7.

Q19 Text Solution:

The Correct option is C.

ROM memory size $= 2^m \times n$

m = no. of address lines

n = no. of data lines

Given, $16K \times 16 = 2^{14} \times 16$

Address lines = 14

Data lines = 16

Q20 Text Solution:

Correct option is C.

This program runs for 100 seconds with multiplications occupying 80% of the runtime. If the goal is to make the program run 4 times faster (i.e., in 25 seconds), we first need to calculate the time spent during the multiplicative operations: $100 \text{ seconds} \times 80\% = 80 \text{ seconds}$. This time needs to come down to $25 \text{ seconds} \times 20\% = 5 \text{ seconds}$ with improved multiplication. So, the speed of multiplication must improve by a factor of $80 \text{ seconds} / 5 \text{ seconds} = 16 \text{ times}$.

Q21 Text Solution:

Option- **B**($m=p^n$)

Explanation:

Steps in Simple Terms:

1. Goal: Calculate p^n (power of p) efficiently.

2. Working:

- m : Starts at 1 and stores the result.
- x : Starts with p and continues to square.
- k : Monitors the remaining exponent, n .

3. Loop Logic: If k is odd, multiply m by x (using that part of the power).

- Square x (set up for the next higher power).
- Reduce the magnitude of the problem by halving (k).

4. At the End: $k = 0$, which terminates the loop.

- m now contains the value of p^n .

Q22 Text Solution:

Option- **C**((1, 2))

Explanation:

Key Points:

1. The structure 'S' size is 4 bytes. (Union 'U' requires two bytes, while 'c' requires one byte and one byte of padding).
2. Pointer Difference (M):
 - p points to 'B[4]', while q points to 'B[5]'.
 - $(q - p = 1)$ (difference between elements of type 'S').
3. Offset for 'c' (N): - 'c' is at offset 2 (after the 2-byte union 'U').
 - $N = 2$.

Q23 Text Solution:

Option- **D**(True, False, False, False)

Explanation:

- **Line 1:** True (because $m1.m == m2.m$)
- **Line 2:** False (because $s1$ and $s2$ are different Good objects)
- **Line 3:** False (due to type mismatch or incorrect comparison)
- **Line 4:** False (because $s1$ and $m2$ are different objects)



Q24 Text Solution:

Option- **D(number of 1's)**

Explanation: The function 'f' calculates the number of 1's in the binary "n" form. This is how it works.

1. Initialization: $b = 0$ counts 1's.

2. Loop: As long as $n \geq 0$:

- $n \& 1$: Determines whether the least significant bit (LSB) is 1.

- If true, b is increased.

- $n \gg= 1$: shifts n one bit to the right while discarding the LSB.

3. Return: Following the loop, b stores the total number of 1s.

Q25 Text Solution:

Option- **A(P only)**

Explanation: In a scripting language such as JavaScript, types are often connected with values rather than variables. JavaScript is a dynamically typed language, hence variables do not have fixed types.

The type is connected with the value contained in the variable rather than the variable itself.

For example:

```
let x = 42;
```

```
x = "Hello";
```

This behavior indicates that in JavaScript, types are associated with values rather than variables. While the `<img>` tag is the standard and most common way to display images on a webpage, it is not the only way. Using CSS backgrounds, SVG, or `<object>/<iframe>` tags are alternative methods for showing images or graphic content without using the `<img>` tag.

Q26 Text Solution:

Option- **A(P and Q only)**

Explanation:

P: XML documents are deemed "Well Formed" if they follow basic syntactic requirements, such as having matching opening and closing tags.

Tags are appropriately nested.

Attribute values are contained in quotations.

Q: "Well Formed" assures proper syntax.

"Valid" signifies that the XML document follows a predefined structure or schema, such as a DTD (Document Type Definition) or an XML Schema. Validation demands that the XML document be Well Formed before being compared to the DTD or schema.

R: The use of `<xml` instead of `<?xml` is a common error. The angle brackets `<` and `>` are used for XML tags, but the XML declaration should always start with `<?` and end with `?>`.

Q27 Text Solution:

Option- **A(450 Kilobytes)**

Explanation: The frame buffer size depends on the resolution and the bits per pixel:

1. Resolution: $640 \times 480 = 307,200$ pixels.

2. Bits per Pixel: 12 bits = 1.5 bytes.

3. Total Memory (in bytes):

$$307,200 \text{ pixels} \times 1.5 \text{ bytes/pixel} = 460,800 \text{ bytes.}$$

4. Convert to KB:

$$460,800 \div 1024 = 450 \text{ KB.}$$

Q28 Text Solution:

Option- **B(Only S1 is true)**

Explanation:

S1: $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ is a 2×2 matrix that reflects (mirrors) only 2D points about the X-axis.

Analysis:

This matrix multiplies the y-coordinate by -1 , leaving the x-coordinate unchanged.

Reflection about the X-axis flips a point's vertical position across the X-axis, i.e., $(x, y) \rightarrow (x, -y)$.

Hence, this matrix correctly reflects 2D points about the X-axis.

Q29 Text Solution:

Option- **A(Both P and Q)**

Explanation:

P: Orthographic transformations keep parallel lines parallel: Orthographic projection is a sort



of parallel projection that projects objects along lines parallel to the viewing direction. Because it does not cause perspective distortion, parallel lines in 3D space stay parallel after the translation.

Q. Orthographic transformations are affine

transformations: An affine transformation maintains points, straight lines, and planes. It may include operations such as translation, scaling, rotation, and shearing.

Orthographic transformations are affine transformations because they preserve point relationships, such as parallelism and distance ratios along parallel lines.

Q30 Text Solution:

Option- **D(the reflected vector and the view vector)**

Explanation: The Phong reflectance model uses the angle between the reflected vector (R) and the view vector (V) to calculate the strength of the specular highlight. This is because specular reflection is determined by how closely the viewer's perspective aligns with the perfect reflection direction of light off the surface.

Key Vectors in the Phong Model:

- 1. Normal vector (N):** Perpendicular to the surface.
- 2. Light vector (L):** Direction from the surface point to the light source.
- 3. View vector (V):** Direction from the surface point to the viewer's eye.
- 4. Reflected vector (R):** Direction of the ideal reflection of the light vector, computed as:

$$R = 2(N \cdot L)N - L$$

Q31 Text Solution:

Option- **B((c) and (d) only)**

Explanation:

- (c) Without this restriction, DRC may produce outcomes that are neither computable nor finite. When you limit DRC to safe expressions, it becomes identical to Relational Algebra.
- (d) Similar to (c), when restricted to safe expressions, Tuple Relational Calculus (TRC) has the same expressive power as Relational Algebra.

When we remove the safety restriction, TRC may provide limitless outcomes, which is undesirable in real applications.

Q32 Text Solution:

Option- **C(Associations with multiplicities)**

Explanation: Foreign keys can represent relationships between entities in relational databases, known as associations with multiplicities.

Primary and foreign keys are used to implement multiplicities, such as one-to-one, one-to-many, and many-to-many.

Q33 Text Solution:

Option- **B((b) and (c))**

Explanation: Set theory includes union, intersection, set difference, Cartesian product, complementation, and power set.

Q34 Text Solution:

Option- **D(R is in 2NF and 3NF)**

Explanation: If R is in BCNF, it meets all conditions of 2NF and 3NF as BCNF is stricter than 3NF.

Q35 Text Solution:

Option- **B(Primary Key)**

Explanation: A primary key is essential for foreign keys to ensure the uniqueness, existence, and reference ability of a table. Only one candidate key is chosen as the primary key.

Q36 Text Solution:

Explanation: A candidate key in a relational database must have uniqueness (P) and irreducibility (Q) properties, ensuring no two rows can have the same value and a minimal subset of attributes.

Q37 Text Solution:

Option- **C((a), (b) and (c))**

Explanation: DML, or Data Manipulation Language, refers to SQL statements that manipulate data stored in a database.

The SELECT statement is used to get data from a database.

The UPDATE command is used to update existing



data in a table.

The DELETE command deletes records from a database table.

Q38 Text Solution:

Option- **C(Map reduce, H Base and Hive)**

Explanation: Hadoop is an ecosystem that includes several technologies for processing and managing large amounts of data.

MapReduce: A Hadoop fundamental component that handles distributed data.

H Base: A NoSQL database built on top of Hadoop Distributed File System (HDFS) to provide real-time read/write access to big datasets.

Hive: A data warehouse program that allows you to query and analyze Hadoop data with SQL-like queries.

Q39 Text Solution:

Option- **B((30,35) and (35,20) Only)**

Explanation:

- To retain the table's referential integrity after deleting 20,40, where 20 is the Primary Key, Any VALUE 20 in the foreign key will be destroyed, and so on. As a result, (35,20) will be erased.
- Further deleting (35,20) where 35 is the Primary Key and to maintain referential integrity any VALUE 35 in the foreign key must be erased, i.e. (30,35).
- Furthermore, no value equal to 30 in the foreign key column is detected.

Q40 Text Solution:

Option- **A(2)**

Explanation:

1. Centroid of C_1 :

Observations in $C_1 = (3, 3), (5, 5), (7, 7)$.

Centroid = $\left(\frac{3+5+7}{3}, \frac{3+5+7}{3} \right) = (5, 5)$.

2. Manhattan Distance for (4, 4):

Distance = $|4 - 5| + |4 - 5| = 1 + 1 = 2$.

Q41 Text Solution:

SSTF Algorithm:

The **Shortest Seek Time First (SSTF)** policy services the request closest to the current head position first. After servicing that request, the head moves to the next closest request.

Execution:

Starting at 50:

Closest request: **34**

Distance: $|50 - 34| = 16$

Move to 34:

Closest request: **20**

Distance: $|34 - 20| = 14$

Move to 20:

Closest request: **19**

Distance: $|20 - 19| = 1$

Move to 19:

Closest request: **15**

Distance: $|19 - 15| = 4$

Move to 15:

Closest request: **10**

Distance: $|15 - 10| = 5$

Move to 10:

Closest request: **7**

Distance: $|10 - 7| = 3$

Move to 7:

Closest request: **6**

Distance: $|7 - 6| = 1$

Move to 6:

Closest request: **4**

Distance: $|6 - 4| = 2$

Move to 4:

Closest request: **2**

Distance: $|4 - 2| = 2$

Move to 2:

Closest request: **73**

Distance: $|2 - 73| = 71$

Total Seek Time:

Summing up all distances:

$16 + 14 + 1 + 4 + 5 + 3 + 1 + 2 + 2 + 71 = 119\text{ms}$

Q42 Text Solution:

List-I	List-II	Explanation
(a) Disk	(iii) File system	The disk is managed by the file system, which



		organizes, stores, and retrieves data efficiently from the disk.
(b) CPU	(i) Thread	The CPU executes threads, which are the smallest units of execution within a process.
(c) Memory	(iv) Virtual address space	Memory is managed using a virtual address space, which provides an abstraction layer for processes to handle memory efficiently and isolate them.
(d) Interrupt	(ii) Signal	An interrupt is a signal sent to the CPU to notify it of an event, such as I/O completion or hardware failure, requiring immediate attention.

Option B is correct.

Q43 Text Solution:

Stack:

NOT shared.

Each thread has its own stack to maintain function calls, local variables, and return addresses. This is necessary because threads operate independently and require their own execution context.

Registers:

NOT shared.

Each thread has its own set of registers, including the program counter and stack pointer, to keep track of its execution state.

Q44 Text Solution:

Option A is Correct

Given Details

- Number of Frames: 3
- Page Reference String: 1, 2, 1, 3, 7, 4, 5, 6, 3, 1

LRU (Least Recently Used) Algorithm

In the LRU algorithm, the page that has not been used for the longest time is replaced when a page fault occurs.

frames	1	2	1	3	7	4	5	6	3	1
f1	1 _L	1 _L	1	1	1 _L	4	4	4 _L	3	3
f2		2	2 _L	2 _L	7	7	7 _L	6	6	6 _L
f3				3	3	3 _L	5	5	5 _L	1

Total Page Faults for LRU: 9

Optimal Page Replacement Algorithm

In the Optimal algorithm, the page that will not be used for the longest time in the future is replaced when a page fault occurs.

frames	1	2	1	3	7	4	5	6	3	1
f1	1	1	1	1	1	1	1	1	1	1
f2		2	2	2 _R	7 _R	4 _R	5 _R	6 _R	6 _R	6 _R
f3				3	3	3	3	3	3	3

Total Page Faults for Optimal: 7

Difference in Page Faults: Difference =

$$\text{Page Faults (LRU)} - \text{Page Faults (Optimal)} = 9 - 7 = 2$$

Q45 Text Solution:

Process P1: Requires 10 units, arrives at time 0.

Process P2: Requires 20 units, arrives at time 2.

Process P3: Requires 30 units, arrives at time 6.

Execution:

Time 0 to 2:

Only **P1** is available.

P1 executes for 2 units. Remaining time for **P1**:

$$10 - 2 = 8.$$

Time 2:

P2 arrives, and it has a remaining time of 20 units.

The scheduler compares **P1** (8 units remaining) with **P2** (20 units remaining). Since **P1** has a shorter remaining time, it continues execution.

Time 6:

P3 arrives with a time requirement of 30 units.

The scheduler compares the remaining times: **P1** (8 units), **P2** (20 units), and **P3** (30 units).



P1 still has the shortest remaining time, so it continues execution.

Time 10:

P1 finishes execution.

The scheduler compares **P2** (20 units) and **P3** (30 units). **P2** is selected.

Time 30:

P2 finishes execution. **P3** starts and runs to completion.

Context Switches:

A **context switch** occurs when the CPU switches from one process to another.

At Time 0: Initial start of **P1** (not counted).

At Time 10: Context switch from **P1** to **P2**.

At Time 30: Context switch from **P2** to **P3**.

Total context switches (excluding at time 0 and the end):

2 switches

Q46 Text Solution:

- **P (wait) operation:** Decrements the semaphore's value by 1.
- **V (signal) operation:** Increments the semaphore's value by 1.

The net change in the semaphore value is:

$$\text{Net Change} = (-20) + 15 = -5$$

The resulting value of the semaphore is:

$$\text{Resulting Value} = 7 + (-5) = 2$$

Q47 Text Solution:

The **minimum number of page frames** that must be allocated to a running process in a virtual memory environment depends on the

Instruction Set Architecture (ISA) of the system.

This is because:

Instruction Set Requirements:

Certain instructions in the ISA may require multiple memory accesses to fetch operands or execute operations. For example:

Some instructions may span multiple pages (e.g., one page for code, another for data).

Indirect addressing or multiple levels of pointers may require additional pages to be loaded.

Q48 Text Solution:

System Resources:

Total tape drives = 6.

Each process may need a maximum of 2 tape drives.

Deadlock Condition: For deadlock to occur, every process must hold some resources and wait for additional resources. To ensure a deadlock-free system, we need to make sure that at least one process can always complete and release its resources.

Safe State: In the worst-case scenario, every process except one holds 1 tape drive (minimum resources), while the remaining tape drives are available for at least one process to complete its execution.

Let the number of processes be n . In the worst case:

- $n-1$ processes hold 1 tape drive each
- Total tape drives held = $n-1$
- Remaining tape drives = $6-(n-1)$

For the system to be deadlock-free, the remaining tape drives are sufficient to meet the maximum need of at least one process (2 tape drives):

$$6-(n-1) \geq 2$$

Simplify the inequality:

$$6-n+1 \geq 2$$

$$7-n \geq 2$$

$$n \leq 5$$

The maximum value of n that ensures the system is deadlock-free is:

5

Q49 Text Solution:

Maximum memory supported: $4 \text{ GB} = 4 \times 2^{30}$ bytes = 2^{32} bytes.

Memory is word addressable: Each word is 2 bytes. So, the total number of addressable words is:

$$\text{Total addressable words} = \frac{\text{Total memory}}{\text{Size of each word}} = \frac{2^{32}}{2} = 2^{31}$$

Determine the size of the address bus

The number of bits required to address 2^{31} words is $\log_2(2^{31})=31$. The **address bus must be at least**



31 bits wide.

Q50 Text Solution:

Option D is Correct

cat *.ch|wc-1 UNIX/Linux pipes will count the number of lines in all the files having .c and .h as their extension in the current working directory.

Q51 Text Solution:

Statement Q:

"A software design is said to be a good design if the components are strongly cohesive and weakly coupled."

Correct.

A good software design adheres to the principles of **high cohesion** and **low coupling**:

High cohesion: Each component should focus on a single responsibility, making it easier to understand, maintain, and test.

Low coupling: Components should have minimal dependencies on each other, allowing for easier modifications and greater reusability.

Q52 Text Solution:

Model (M):

The **Model** is responsible for managing the application's **business logic and domain objects**. It handles the data, logic, and rules of the application.

View (V):

The **View** is responsible for presenting data to the user. It handles the **user interface (UI)** and displays the output generated by the application.

Controller (C):

The **Controller** acts as an intermediary between the Model and the View.

Option C (business logic and domain objects):

Correct. The Model is designed to handle business logic and manage the domain objects.

Q53 Text Solution:

Statement P:

"A minimal test set that achieves 100% path coverage will also achieve 100% statement coverage."

Correct.

Path coverage is a stricter testing criterion than

statement coverage. Achieving 100% path coverage ensures that every possible path through the code is tested, which includes every statement being executed at least once.

Q: A minimal test set that achieves 100% path coverage will generally detect more faults than one that achieves 100% statement coverage.

This statement is correct.

Path coverage tests multiple paths through the program, which can reveal defects that arise from specific combinations of statements. On the other hand, statement coverage only ensures that every statement is executed at least once, but does not guarantee that all paths (and potential interactions between statements) are covered. As a result, achieving 100% path coverage is generally more effective at detecting faults than 100% statement coverage.

R: A minimal test set that achieves 100% statement coverage will generally detect more faults than one that achieves 100% branch coverage.

This statement is NOT correct.

Branch coverage ensures that every possible branch (decision point) in the program is taken at least once, which typically provides more thorough testing than statement coverage, which only ensures that each statement is executed at least once, regardless of the branches. In many cases, branch coverage will detect more faults because it takes into account the behavior of the program under different decision outcomes, whereas statement coverage does not consider how combinations of statements might lead to errors. Therefore, a test set achieving 100% branch coverage will generally detect more faults than one achieving 100% statement coverage.

Q54 Text Solution:

Software reuse refers to the practice of leveraging existing software components, artifacts, or knowledge to build new software applications. It helps save development time, reduce costs, and improve software quality by reusing reliable and tested components.



Option B: the process of using existing software artifacts and knowledge to build new software

Artifacts can include code libraries, frameworks, designs, documentation, or even knowledge about best practices.

Q55 Text Solution:

Git is a **Distributed Version Control System (DVCS)** designed to track changes in source code during software development. It allows multiple developers to work on the same project simultaneously, manage code revisions, and collaborate efficiently.

- **Version Control:** Tracks changes in files and allows users to revert to previous versions.
- **Distributed Nature:** Every developer has a complete copy of the repository, ensuring redundancy and enabling offline work.
- **Branching and Merging:** Supports non-linear development workflows.

Option C is correct.

Q56 Text Solution:

Integrity = $1 - (\text{Threat Probability} \times (1 - \text{Likelihood of Repelling the Attack}))$

Threat Probability = 25% = 0.25

Likelihood of Repelling an Attack = 30% = 0.30

Step-by-Step Calculation:

Compute $1 - \text{Likelihood of Repelling the Attack}$:
 $1 - 0.30 = 0.70$

Multiply the **Threat Probability** with the result from Step 1:

$0.25 \times 0.70 = 0.175$

Subtract the result from Step 2 from 1 to compute the integrity:

Integrity = $1 - 0.175 = 0.825$

Option D is correct.

Q57 Text Solution:

Option D is Correct

List-I (Software Process Models)	List-II (Software Systems)	Explanation
(a) Waterfall Model	(iv) Automate the manual system for student record maintenance in a school	The Waterfall model is suitable for systems with clear and stable requirements, like a student record system.
(b) Incremental Development	(i) e-business software that starts with only basic functionalities and then moves on to more advanced features	Incremental development delivers the product in stages, making it ideal for evolving e-business platforms.
(c) Prototyping	(iii) A virtual reality system for simulating vehicle navigation in a highway	Prototyping helps refine requirements for complex and interactive systems like virtual reality applications.
(d) RAD (Rapid Application Development)	(ii) An inventory control system for a supermarket to be developed within three months	RAD emphasizes quick development and delivery, suitable for short deadlines like a supermarket inventory system.

Q58 Text Solution:

Adaptive maintenance refers to modifications made to a software product to ensure that it continues to function properly in a changing environment. This includes adapting the software to new platforms, and operating systems.

Option B: When the customers need the product to run on new platforms

Adaptive maintenance ensures that the software



can operate on new hardware or software platforms that emerge.

Q59 Text Solution:

Risk monitoring is an essential part of risk management in software project tracking. It ensures that the project team proactively tracks and handles risks.

P: To assess whether predicted risks do, in fact, occur

This helps in understanding how well the initial risk assessment was done and whether those risks affect the project.

Q: To ensure that risk aversion steps defined for the risk are being properly applied

Risk mitigation is critical to handling identified risks. Risk monitoring ensures these steps are implemented correctly and effectively to minimize the potential impact of risks.

R: To collect information that can be used for future risk analysis

Risk monitoring involves gathering data on how risks are handled, which is valuable for improving risk management practices in future projects.

Q60 Text Solution:

Software validation ensures that the final product meets the user's requirements and fulfills its intended purpose. It checks "are we building the right product?" by verifying whether the **detailed specifications align with the user's actual needs**.

Option C: detailed specifications and user requirements

It ensures the software aligns with the documented needs of the user.

Q61 Text Solution:

Option B is Correct

List-I (Algorithm)	List-II (Complexity)	Explanation
(a) Prim's Algorithm	(iii) $O(E \log V)$	Prim's algorithm implemented with a binary heap and adjacency list achieves $O(E \log V)$.

List-I (Algorithm)	List-II (Complexity)	Explanation
(b) Dijkstra's Algorithm	(iv) $O(V^2)$	Dijkstra's algorithm for dense graphs (using an adjacency matrix) takes $O(V^2)$.
(c) Floyd-Warshall Algorithm	(i) $O(V^3)$	Floyd-Warshall algorithm for all-pairs shortest paths has a cubic complexity for dense graphs.
(d) Edmonds-Karp Algorithm	(ii) $O(VE^2)$	Edmonds-Karp algorithm (Ford-Fulkerson implementation) takes $O(VE^2)$.

Q62 Text Solution:

Option D is Correct

Insertion Sort: Sorts in place with worst case time and space complexity of $O(n^2)$ and $O(1)$ respectively.

Merge Sort: Sorts out of place with worst case time and space complexity of $O(n \log n)$ and $O(n)$ respectively.

Q63 Text Solution:

Option D is Correct

To calculate $\phi(45)$ using Euler's Phi function, let's proceed step by step.

Euler's Phi Function Formula: $\phi(n) = n \prod \left(1 - \frac{1}{p}\right)$

Where p runs over all prime factors of n.

Prime Factorization of $45 = 3^2 \times 5$. So, Prime factors of 45: $p=3, 5$

Apply the Formula:

$$\phi(45) = 45 \times \left(1 - \frac{1}{3}\right) \times \left(1 - \frac{1}{5}\right)$$

$$\phi(45) = 45 \times \frac{2}{3} \times \frac{4}{5} = 24$$

Q64 Text Solution:

Option B is Correct

1. $O(\lg n)$: Too fast to be achievable for sorting n elements. Not correct for sorting problems.
2. $O(n)$: Achievable using Counting Sort or Radix Sort when the range is manageable (like n^2-1).
3. $O(n \lg n)$: Time complexity of comparison-based algorithms like Merge Sort, Quick Sort, or Heap Sort. Not optimal here because



specialized algorithms (Counting Sort, Radix Sort) outperform comparison-based sorts for this problem.

4. $\Theta(n^2)$: Too slow and unnecessary for this input range and problem constraints.

Q65 Text Solution:

Option C is Correct

Depth-First Search (DFS) is a fundamental algorithm used to explore graphs and has numerous applications in graph theory. Among its applications, Topological Sort and Strongly Connected Components (SCCs) are key, and both rely on DFS as their core algorithm. Let's analyze:

1. Topological Sort

Topological sort is a linear ordering of vertices in a Directed Acyclic Graph (DAG) such that for every directed edge $u \rightarrow v$, vertex u comes before v in the ordering. Role of DFS:

- DFS is used to explore the graph and determine the finishing times of each node.
- Nodes are added to the stack in reverse order of their finishing times during DFS traversal, ensuring a valid topological order.

2. Strongly Connected Components (SCCs)

A strongly connected component of a directed graph is a maximal subgraph where every vertex is reachable from every other vertex in the same component. Role of DFS:

- DFS is used twice in Kosaraju's Algorithm or Tarjan's Algorithm to find SCCs:
 - First DFS: Determines the finishing times of vertices.
 - Second DFS (on the transposed graph): Explores the graph in the reverse order of finishing times to identify SCCs.

Both Topological Sort and Strongly Connected Components are significant applications of DFS:

- Topological Sort is used for linear ordering in a DAG.

- Strongly Connected Components help identify cohesive substructures in directed graphs.

Hence, both rely on DFS for their computation.

Q66 Text Solution:

Option C is Correct

Given that, $h(k, i) = (h_1(k) + ih_2(k)) \bmod m$

where $h_1(k) = k \bmod m$, $h_2(k) = 1 + (k \bmod n)$ where $n = m - 1$, $m = 701$

Find for $k = 123456$

Now, $h_1(k) = 123456 \bmod 701 = 80$

And, $h_2(k) = 1 + (123456 \bmod 700) = 1 + 256 = 257$

1st probe: when $i = 1$,

$h(k, i) = h_1(k) + ih_2(k) = h(k, 1)$

$= h_1(k) + h_2(k) = 80 + 257 = 337$

2nd probe: when $i = 2$,

$h(k, 2) = h_1(k) + 2h_2(k) = 80 + 2 \times 257 = 80 + 514 = 594$,

So, difference between first two probes $= 594 - 337 = 257$.

Q67 Text Solution:

Option C is Correct

Here, CO-NP is A language $L \in \text{CO-NP}$ if $L' \in \text{NP}$, where L' is the complement of L . Intuitively, CO-NP contains decision problems whose complements are verifiable in polynomial time. And, P is the set of decision problems that can be solved in polynomial time.

Also, NP is the set of decision problems whose solutions can be verified in polynomial time.

Now based on this analysing the statements, $S1: P \subseteq \text{CO-NP}$

Any problem in P is solvable in polynomial time. If $L \in P$, then both L and its complement L' can be solved in polynomial time, implying $L' \in P \subseteq \text{NP}$.

Hence, $L \in \text{CO-NP}$ because $L' \in \text{NP}$. So, $S1$ is True.

$S2: \text{If } \text{NP} \neq \text{CO-NP}, \text{ then } P \neq \text{NP}$

If $\text{NP} \neq \text{CO-NP}$, it implies that there exists at least one problem in NP whose complement is not in NP, making CO-NP strictly larger than P. And If $P = \text{NP}$, then $P = \text{CO-NP}$ because both classes become equal. Therefore, for $\text{NP} \neq \text{CO-NP}$, it must hold that $P \neq \text{NP}$. So, $S2$ is also True.

Q68 Text Solution:



Option A is Correct

Only S1, because both dynamic and greedy finds optimal solution but only dynamic programming computes the value by following a Bottom up approach.

Q69 Text Solution:

Option C is Correct

A flow network is a directed graph $G=(V,E)$ where:

1. Each edge (u,v) has a capacity $c(u,v) \geq 0$.
2. A flow $f: V \times V \rightarrow R$ is assigned to each edge (u,v) , satisfying specific properties.

P1: $f(u,v) = -f(v,u)$ for all $u,v \in V$

This property ensures antisymmetry of the flow:

- The flow from u to v is the negative of the flow from v to u .
- For example, if $f(u,v)=5$, then $f(v,u)=-5$.

This is a standard property in flow networks because flow in the reverse direction cancels out the original flow. Hence, P1 is True.

P2: $\sum_{v \in V} f(u,v) = 0$ for all $u \in V$

This property is called the flow conservation property. At every vertex u , except the source and sink, the total flow entering u equals the total flow leaving u . This ensures that no flow is "lost" or "created" at intermediate vertices. This property is fundamental to flow networks and is always true. Hence, P2 is also True.

Q70 Text Solution:

Option A is Correct

S1: For any integer $n > 1$, $a^{\phi(n)} \equiv 1 \pmod{n}$ for all $a \in \mathbb{Z}_n^*$, where $\phi(n)$ is Euler's phi function.

This statement is a direct consequence of Euler's Theorem:

Euler's theorem states that for any integer $n > 1$ and any $a \in \mathbb{Z}_n^*$ (i.e., a is coprime to n): $a^{\phi(n)} \equiv 1 \pmod{n}$.

This result follows from the fact that $\phi(n)$ counts the number of integers less than n that are coprime to n .

This theorem is always true under the given conditions.

Hence, S1 is true.

S2: If p is prime, then $a^p \equiv 1 \pmod{p}$ for all $a \in \mathbb{Z}_p^*$.

$a \in \mathbb{Z}_p^*$ implies that a is coprime to p , which is trivially true since p is prime. However, the statement $a^p \equiv 1 \pmod{p}$ is not true for all such a .

By Fermat's Little Theorem, we know: $a^{p-1} \equiv 1 \pmod{p}$.

Fermat's theorem does not guarantee $a^p \equiv 1 \pmod{p}$ (this would hold only if $a \equiv 1 \pmod{p}$).

Hence, S2 is false.

Q71 Text Solution:

Option C is Correct

A symbol table is a data structure used by a compiler to store and manage information about variables, functions, objects, and other program entities. It keeps track of:

- Variable names and their associated attributes (e.g., type, scope, memory location).
- Function names, their return types, and parameter lists.
- Constants and their values.
- Objects and classes in object-oriented languages.

Q72 Text Solution:

Option B is Correct

The given expression is:

$(i*j) + (e+f) * (a*b+c)$

We are tasked with determining how many operations are required to translate this expression into quadruple representation. Each arithmetic operation $(+, *)$ will be considered a separate operation in the quadruple representation.

Quadruple Representation:

Step	Operator	Operand 1	Operand 2	Result
1	*	I	J	t1
2	+	e	F	t2
3	*	a	b	t3
4	+	t3	C	t4
5	*	t2	t4	t5
6	+	t1	t5	t6



Where,

Temporary Variable	Meaning
t1	Result of $i*j$
t2	Result of $e+f$
t3	Result of $a*b$
t4	Result of $t3+c$
t5	Result of $t2*t4$
t6	Final result: $t1+t5$

Making Observations from the Table there are 6 Operations.

Q73 Text Solution:

Option A is Correct

A. Constant folding is a compiler optimization technique where constant expressions are evaluated and replaced with their computed values during the compile time, instead of being evaluated at runtime. This is part of the optimization process to reduce runtime overhead and improve program execution speed.

In Context of the Given Expression: The given expression is: 4×2.14

- The compiler identifies that both 4 and 2.14 are constants.
- It computes the value of this expression at compile time: $4 \times 2.14 = 8.56$
- The result 8.56 is substituted into the code, replacing the original expression 4×2.14 .

Q74 Text Solution:

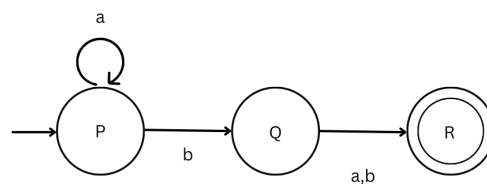
Option D is Correct

SR parser is a bottom up parser. It consists of input buffer, stack, parse table. Where Input buffer is used for storing strings to be parsed, stack is used for holding grammar symbols and parse table is used for parsing the string.

Q75 Text Solution:

Option C is Correct.

As can be observed with the given Construction as per the Equation, there are 3 States.



Q76 Text Solution:

Option B is Correct

Statement A: (ii)

The complement of recursive language is recursive which is union by Recursively Enumerable Language which gives Recursively Enumerable Language.

Statement B: (iii)

Complement of Context-Free Language is REC language which is union by REC, REC is closed under union so result is REC.

Statement C: (i)

Kleen closure of REG language is REG, intersection with CFL gives CFL.

Q77 Text Solution:

Option C is Correct

To decide whether a context-free language L is finite, we need to check if there are only a finite number of strings in L . The correct method involves the following considerations:

- If L is finite, it must not contain any infinite derivations or cycles in its derivation process.
- By removing unnecessary productions and non-productive variables (creating a non-redundant CFG) and considering only the derivations that do not involve generating the null string (ϵ), we can determine whether the language is finite.

The correct approach is by constructing non-redundant CFG G in CNF generating language $L \setminus \{\epsilon\}$



1. Non-redundant CFG: The CFG is stripped of redundant productions and variables that do not contribute to generating strings in L .
2. CNF (Chomsky Normal Form): Converting the CFG into CNF ensures that every production is simplified, which makes analysis (like detecting cycles or infinite derivations) easier.
3. Excluding ϵ : Removing the null string (ϵ) ensures that we focus only on derivations producing actual strings, and this avoids trivial infinite derivations involving ϵ .

Q78 Text Solution:

Option D is Correct

A smallest string produced by this grammar will be 'abba'. So, statement (a) and (b) are not Incorrect. Whereas statement (c) is correct and every string produced by this grammar have at least two a's always.

Statement (d) is also correct because every string produced by the grammar have b's in multiple of 2. $L = \{ abba, bbaabbabba, \dots \}$

Q79 Text Solution:

Option C is Correct

(a): $L(G) = \phi$

It means that to Determine if a context-free grammar G generates an empty language. The emptiness problem for context-free languages (CFLs) is decidable. It is possible to algorithmically analyze whether any derivation in G leads to a terminal string by marking useful variables and checking their reachability. Hence, Statement a is Decidable.

(b): Checking $L(M) \subseteq L(M_1) \cup L(M_2)$

The problem is decidable if we assume $L(M_2) = \Sigma^*$ because any $L(M)$ will automatically be a subset of $L(M_2)$. Therefore, Check whether M_1 is a valid Turing machine, which is decidable. And there is No need to verify the subset relationship as it is trivially true under the assumption.

If $L(M_2) = \Sigma^*$, $L(M_1) \cup L(M_2) = \Sigma^*$, and $L(M) \subseteq \Sigma^*$ is always true for any Turing machine M . Validity of a Turing machine can be checked (decidable).

The assumption makes the subset condition

redundant, reducing the problem to verifying M_1 's validity.

Hence, This Statement b is Decidable under the provided assumption.

(c): Checking M accepts w using at most $2^{|w|}$ tape cells

The problem is recursive because we can compute the number of possible configurations $\alpha = m \cdot k^{2r} \cdot 2r$, where:

- m : Number of states in M ,
- k : Size of the tape alphabet,
- $r = |w|$: Length of the input.

If M runs for more than $\alpha + 1$ steps, it must revisit a previous configuration (by the pigeonhole principle), entering an infinite loop. A Turing machine M^* can decide the problem by simulating M for $\alpha + 1$ steps and verifying tape usage.

M^* halts in all cases as it Accepts w if M accepts within the tape bound. Rejects w if M uses more than $2^{|w|}$ cells or exceeds $\alpha + 1$ steps. Therefore, this statement c is decidable.

Q80 Text Solution:

Option C is Correct

It is recursive language then Turing Machine may accept or reject but it will definitely halt at non-final state.

Q81 Text Solution:

Option D is Correct

Fully Connected Network Topology: A fully connected network is one where every node is directly connected to every other node. The number of direct links in such a network is equal to the number of unique pairs of nodes. To determine the number of direct links:

1. For n nodes, each node can connect to $n-1$ other nodes.
2. However, this counts each link twice (once from $A \rightarrow B$ and once from $B \rightarrow A$).
3. To count each link only once, divide by 2:

$$\text{Number of Links} = \frac{n(n-1)}{2}$$



Q82 Text Solution:

Explanation:

RARP allows a client (host) to send a broadcast message containing its MAC address and request an IP address.

The Final Answer is: C

Q83 Text Solution:

Explanation:

Distance = 800 km

Speed of signal in medium = 200000 km/second.

$$\text{Propagation Delay} = \frac{\text{Distance}}{\text{Speed of Signal in Medium}} = \frac{800 \text{ km}}{200000 \text{ km/second}} = 0.004 \text{ Seconds} = 4 \text{ milliseconds}$$

The Final Answer is: D

Q84 Text Solution:

Explanation:

Total subnet $\rightarrow 500 \text{ subnet} \approx 2^n \rightarrow n = 9 \text{ bit}$

(Using both side logarithms)

Host address = $2^n - 2 \rightarrow 100 = 2^n \rightarrow 102 = 2^n \rightarrow n = 7 \text{ bit}$

Class B = Fixed Network Id $\rightarrow 16 \text{ bit}$.

Subnet id = 9 bit

(Network ID and Subnet ID must be 1 and Host ID must be 0)

Network id = 11111111 11111111 (16 bit)

Subnet id = 11111111 1 (9 bit)

Host id = 00000000 (7 bit)

Subnet mask: 11111111 11111111 11111111

10000000 $\rightarrow 255.255.255.128$

The Final Answer is: 255.255.255.128

Q85 Text Solution:

Explanation:

The loopback (IP) address is a member of the Class A network. It falls within the range of 127.0.0.0 to 127.255.255.255.

A loopback (IP) address is used to send a packet from the host to itself.

The Final Answer is: P-False; Q-True

Q86 Text Solution:

Explanation:

In the TCP/IP model, encryption and decryption are functions of the Application layer.

The Final Answer is: D

Q87 Text Solution:

Explanation:

In the TCP/IP model, the data transmission is referred to as a frame at the Link layer, also known as the Data Link layer in the OSI model. At the Network layer of the TCP/IP model, data is sent as a packet, not as a segment.

The Final Answer is: A

Q88 Text Solution:

Explanation:

IPv4 address = 32 bit

Total Number of IP addresses = 2^{32}

No of the IP address in Class C = 2^{29}

$$\text{Percentage of Class C addresses} = \frac{2^{29}}{2^{32}} \times 100 = 2^{-3} \times 100 = \frac{100}{8} = 12.5\%$$

The Final Answer is: 12.5%

Q89 Text Solution:

Explanation:

The ability to inject packets into the Internet with a false source address is known as IP spoofing.

The Final Answer is: D

Q90 Text Solution:

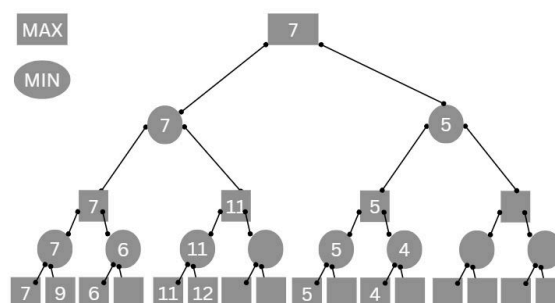
Explanation:

The RSA encryption algorithm also works in reverse, that is, you can encrypt a message with the private key and decrypt it using the public key. This property is used in digital signatures.

The Final Answer is: B

Q91 Text Solution:

Explanation:



The Final Answer is: 7

Q92 Text Solution:

Explanation:



	List I		List II
(a)	Greedy best-first	(ii)	Minimal h(p)
(b)	Lowest cost-first	(iii)	Minimal cost (p)
(c)	A* Algorithm	(i)	Minimal cost (p) + h(p)

The Final Answer is: (a)-(ii); (b)-(iii); (c)-(i)

Q93 Text Solution:

Explanation:

The STRIPS representation is action-centric.

STRIPS stands for Stanford Research Institute

Problem Solver

It is specially used in specifically in the field of planning.

The Final Answer is: B

Q94 Text Solution:

Explanation:

The duality condition ensures that $t(x,y)$ using (fuzzy AND) and $s(x,y)$ using (Fuzzy OR) are opposites under complemented inputs.

$$t(x,y) = 1 - s(1-x, 1-y)$$

The Final Answer is: Option C

Q95 Text Solution:

Explanation:

In fuzzy set theory, α -cuts (A_α) represent subsets of elements with membership values greater than or equal to α .

For $\alpha_1 < \alpha_2$:

A_{α_1} (with the smaller threshold α_1) will contain all elements of A_{α_2} , and possibly more.

This implies: $A_{\alpha_1} \supseteq A_{\alpha_2}$.

The Final Answer is: Option A

Q96 Text Solution:

Explanation:

In fuzzy logic, defuzzification converts fuzzy outputs into crisp values. The methods used are as follows:

M1: Mean of Maximum (MOM) Averages the value at the maximum membership degree.

M2: Centre of Area (COA): finds the centroid of the fuzzy set.

M3: Height Method: Picking the crisp value at the fuzzy set's peak.

The Final Answer is: D

Q97 Text Solution:

Explanation:

All four components—evolution, selection, reproduction, and mutation—are found in genetic algorithms.

Evolution: This is the overall process of adapting a population of solutions across generations.

Selection: The process of selecting individuals from the population based on their fitness or performance.

Reproduction: Mechanisms of creating a new individual (offspring), such as crossover (recombination) and mutation.

Mutation refers to the random change of organisms to introduce variation in a population.

The Final Answers is: (a), (b), (c), and (d)

Q98 Text Solution:

Explanation:

Self-organizing feature maps are a very popular example of an unsupervised neural network.

Self-organizing feature maps high-dimensional input data into a lower-dimensional grid without labeled data; they also learn by similarity between the input data.

The Final Answer is: D

Q99 Text Solution:

Explanation:

$$f(x) = \frac{1}{1 + e^{-2x}}$$

To solve for the derivative of the Sigmoid function:

$$\begin{aligned} \Rightarrow f'(x) &= \frac{d}{dx} \left(\frac{1}{1 + e^{-2x}} \right) \\ &= \frac{-1 \cdot \frac{d}{dx} (1 + e^{-2x})}{(1 + e^{-2x})^2} \end{aligned}$$

Now, differentiate the inside function $1 + e^{-2x}$

$$\frac{d}{dx} \left(\frac{1}{1 + e^{-2x}} \right) = -2e^{-2x}$$

$$\text{Thus, the derivative is: } f'(x) = \frac{2 \cdot e^{-2x}}{(1 + e^{-2x})^2}$$

Put the value of $x=0$ (Given)



$$f'(x) = \frac{2 \cdot e^0}{(1+e^0)^2}$$

$$\Rightarrow f'(x) = \frac{2 \cdot e^0}{(1+e^0)^2} = \frac{2 \cdot 1}{(1+1)^2} = \frac{2}{(2)^2} = \frac{2}{4}$$

$$= \frac{1}{2}.$$

The Final Answer is: B

Q100 Text Solution:

Explanation:

In Reinforcement Learning, the task for the agent is described as a Markov decision process, such that it is aware of the states and possible actions but must figure out which action brings the best outcome through trial and error.

The Final Answer is:C



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